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The SPoRC Stock Assessment Package: A Generalized Next-Generation Platform to Assess Spatial, Age and Sex-Structured Populations

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ABSTRACT

Fisheries managers increasingly rely on stock assessment models to provide sustainable management advice, which must balance complex and competing objectives. Although assessment models necessitate simplifying assumptions, there is increasing recognition to improve the integration of realistic biological and fishery dynamics. While ongoing and future development of increasingly complex stock assessments requires increased capacity (i.e., time, training, and personnel), there are concerns that the growing demand for assessment advice cannot be adequately fulfilled under current resources. To address these needs, we present the Stochastic Population over Regional Components (SPoRC) model, a modern integrated analysis platform that is generalized, modular and developed in a user-friendly programming language (R with Template Model Builder, RTMB). Development of SPoRC emphasized consideration of age, sex, and spatial dynamics at the outset, while ensuring the ability to integrate diverse data sources (e.g., catch, abundance indices, age/length compositions and tagging). Moreover, state-space specification of time-varying processes through random effects and robust simulation capabilities were primary design features. Here, we highlight SPoRC's utility in cohesively implementing single-and multi-region applications, its ability to estimate time-varying processes as random effects, and the potential to conduct closed-loop simulations for evaluating the robustness of management procedures. By developing a generalized assessment platform in a user-friendly R package, we suggest that SPoRC reduces barriers to implementation of next-generation stock assessment models and facilitates team-based model development, which will ultimately help improve the capacity to implement stock assessments globally.

1 | Introduction

Fisheries management systems must reconcile a suite of competing objectives, spanning biological, socio-economic, political and environmental contexts (Hilborn 2007). Often, managers are tasked with implementing catch advice that are biologically sustainable, economically viable and socially equitable, while explicitly accounting for uncertainty in environmental conditions (Free et al. 2025). Stock assessment models are often the first step in forming the scientific basis of fisheries management decisions. Estimates from these models reflect fish population dynamics for determining the population status and provide guidance for near- and long-term management policies. These models are often complex, and while efforts to improve the transparency of catch advice have generally been successful, evaluations remain limited to an insular guild of experts (Hilborn 2003). The demand for stock assessment expertise continues to grow as management agencies are tasked with providing quantitative advice (e.g., total allowable catch) for an increasing number of species (Hilborn and Ovando 2014). Although assessment capacity is improving, limited training opportunities in quantitative population dynamics and the complexity of standard assessment software platforms can potentially impede meeting management needs (Berkson et al. 2009; Maunder et al. 2025). Thus, strengthening capacity within the stock assessment enterprise through targeted training, expanded opportunities for mentorship and cross-collaboration and, particularly, improved accessibility of assessment tools remains an ongoing need (Dichmont et al. 2025; Lynch et al. 2018).

Contemporary stock assessment models follow an integrated analysis paradigm, where multiple data sources are integrated within a unified framework to model population dynamics (Maunder and Punt 2013). Often, contemporary models are implemented with generalized, extensible platforms, which help identify and implement stock assessment good practices, facilitate open and reproducible science, and reduce coding errors (Dichmont et al. 2021). The primary integrated assessment packages that have been used for much of the 2000s include, but are not limited to: SS3 (Stock Synthesis 3; Methot and Wetzel 2013), CASAL2 (C++ algorithmic stock assessment library; Doonan et al. 2016), GADGET (Globally applicable Area Disaggregated General Ecosystem Toolbox; Begley and Howell 2004), and MULTIFAN-CL (MULTIple length Frequency ANanlyis—Catch at Length; Fournier et al. 1998). Some of these packages were developed in ADMB (Automatic Differentiation Model Builder; Fournier et al. 2012) or directly in C++ to leverage automatic differentiation techniques for efficient model estimation. However, many of these packages face maintenance and support limitations. In recent years, TMB (Kristensen et al. 2016) has emerged as a successor to previous implementations of non-linear function minimizers leveraging automatic differentiation. The origins of TMB closely follow the characteristics of ADMB, but uses faster and more efficient external coding libraries, with important improvements on the ability to efficiently estimate random effects (i.e., unobserved processes). Increasing adoption of TMB is evident through several stock assessment platforms, such as SAM (State-space Assessment Model; Nielsen and Berg 2014), WHAM (Woods Hole Assessment Model; Miller et al. 2025; Stock and Miller 2021), and FIMS (Fisheries Integrated Modelling System; FIMS Implementation Team 2022). However, a persistent challenge is that both ADMB and TMB are unfamiliar to many fisheries scientists and can hinder underlying code development, collaboration, and knowledge transfer. Recently, the development of RTMB (R bindings for TMB) aimed to help lower barriers to implementation of statistical state-space models (e.g., stock assessments) and associated code development by allowing users to access C++ template functionality from R, a language widely utilized in fisheries science. However, to our knowledge, no fully generalized and publicly available stock assessment platforms have been developed in RTMB.

Debate continues over the optimal design and requirements of next generation of stock assessment platforms (Punt et al. 2020; Dichmont et al. 2021; Berger et al. 2024). For instance, future stock assessment platforms should be modular and flexible, keep pace with advances in stock assessment methods, integrate novel data sources, and be able to readily address contemporary fisheries management challenges (Punt et al. 2020). Throughout, the term 'modular' is intended to reflect a design in which model components are discrete, loosely coupled, and can be developed in a generally independent manner. In general, a unified system spanning model development, testing, estimation and projections will be necessary to improve transparency, ensure consistent assumptions and ensure limited potential for user-error. Moreover, reproducible assessment workflow tools, including model diagnostics, visualizations of model outputs and simulations to test model and management robustness, will also be critical components of a next-generation assessment platform (Punt et al. 2020, 2016; Taylor et al. 2021). In particular, direct integration of feedback simulation capabilities (i.e., management strategy evaluations; MSEs), within a unified workflow would streamline the evaluation of assessment-management performance and reduce reliance on linking multiple software applications across stages of the assessment process. Thus, through the combination of a reproducible and transparent workflow, along with a more approachable programming language (e.g., R), it is anticipated that broader interdisciplinary participation in stock assessment can facilitate the incorporation of ecology, oceanography, and related disciplines to inform robust model assumptions (Baetscher et al. 2025; Kuparinen et al. 2012).

Ultimately, next generation platforms must remain adaptable, allowing integration of more complex models that capture advances in biological understanding of species, and readily respond to evolving ecosystem and enviornmental conditions. For example, population processes are increasingly non-stationary, and next-generation assessment models must be capable of adequately representing time-varying dynamics (e.g., through state-space methods; Nielsen and Berg 2014; Stock and Miller 2021). Likewise, because ecosystem interactions are inherently spatial, next-generation platforms should accommodate a range of population and spatial structures to ensure that management actions align with ecological scales, even as species distributions shift beyond traditional jurisdictions (Karp et al. 2019). As such, easing the implementation of spatial stock assessment models, while simulatenously enabling a cohesive transition to spatially-aggregated approaches within a single platform, will be a desirable feature (Berger et al. 2024). Moreover, there is also growing recognition that a variety of fish species can exhibit sex-specific dynamics, and that stock assessment models must be capable of accomodating these dynamics to minimize biases (Cheng, Goethel, et al. 2025). Taken together, these considerations underscore the necessity for adatapble modelling platforms that can accomdoate a wide range of biological and environmental dynamics, thereby enabling the exploration of multiple hypotheses and structural assumptions within the assessment process.

To meet the broad and diverse needs of a next generation stock assessment platform, we developed the Stochastic Population over Regional Components (SPoRC) model using RTMB (Kristensen et al. 2016), a user-friendly R package (R Core Team 2021). SPoRC supports single- and multi-region models, time-, age- and sex-varying processes, and integrated analyses at the scale of data collection. We demonstrate its utility with three case studies that highlight spatial modelling, random effects and closed-loop simulations. As a modular platform built entirely in RTMB, SPoRC aims to provide an accessible and flexible tool for contemporary assessment challenges.

2 | Methods

We first describe the design principles and structure of the SPoRC platform, then summarize its main model components. We illustrate SPoRC with three case studies: (1) a sex-structured spatial model for a long-lived, highly mobile species (Alaska sablefish, *Anoplopoma fimbria*), (2) a non-spatial model with time-varying processes for a moderately short-lived species (Eastern Bering Sea walleye pollock, *Gadus chalcogrammus*) and (3) closed-loop simulations to evaluate alternative management procedures for a long-lived demersal species (Gulf of Alaska dusky rockfish, *Sebastes variabilis*). A detailed mathematical description is provided in Data S1. The open-source code repository can be found at https://github.com/chengmatt/SPoRC, and the associated package web page with additional code examples is available at https://chengmatt.github.io/SPoRC/.

2.1 | Conception of SPoRC and Design Principles

SPoRC was initially developed to support and eventually replace the operational and research applications of the Alaska sablefish assessment, which was previously implemented using a bespoke ADMB model (Fournier et al. 2012) In particular, the bespoke model had become difficult to update and extend, given the array of complexities that needed to be addressed for Alaska sablefish (i.e., time-varying processes, fleet-structure, sex-structure, and spatial structure; Cheng et al. 2023; Cheng, Goethel, and Cunningham 2024; Cheng, Goethel, Hulson, et al. 2024). Thus, SPoRC was implemented in RTMB (Kristensen et al. 2016) which combines automatic differentiation and efficient random-effects estimation with a familiar R interface. Moreover, SPoRC was designed to ensure close correspondence with the original sablefish ADMB model, and applications translating the bespoke sablefish model to SPoRC demonstrated remarkable similarity, with differences in model estimates $<0.1\%$, effectively bridging the two platforms (Goethel and Cheng 2025; Goethel and Cheng 2024). Starting in 2026, SPoRC will be used operationally for conducting Alaska sablefish stock assessments. From the outset, SPoRC was developed to ensure flexibility for defining model partitions (e.g., regions, ages, sexes) and support rapid exploration of alternative model configurations. The framework emphasized: (1) modular coding, enabling user-defined functions for key processes (e.g., selectivity), (2) compatibility with next-generation features such as random effects, simulation testing, and flexible specification of spatial structures (Berger et al. 2024; Punt et al. 2020) and (3) integration with existing R packages for data processing and model visualizations. Importantly, detailed package vignettes were developed to further enhance the usability of SPoRC and to engage additional disciplines beyond stock assessment experts. Users should be familiar with basic R and RTMB concepts (data and parameter lists, mapping structures and non-linear minimization).

2.2 | Model Structure and Key Features

The SPoRC platform is comprised of several generalized model partitions, including region (spatial or single-region), year, age, sex (single- or two-sex) and fleet (fishery and survey; any number can be accommodated per region). In general, SPoRC operates on an annual basis where population dynamics are applied in a sequential order: recruitment occurs first, followed by discrete Markovian movement, continuous yearly mortality, and then age advancement (Data S1). While the existing features of SPoRC do not currently explicitly account for seasonality, seasonal dynamics can be approximated by treating years as time steps, which is sometimes assumed in tuna models (Goethel et al. 2024). Below, we summarize the model's implementation, population structure, connectivity, fleet processes, data integration, estimation, projections, diagnostics and simulation features.

2.2.1 | R Package Structure and Model Implementation

SPoRC is structured to facilitate the configuration of a stock assessment model through a series of modular helper functions. These functions aid in the definition of model structure, specify parameterizations for key population and observation processes (e.g., recruitment, fishery selectivity) and organize data inputs into a consistent format (Table 1). In general, data are provided as R objects in the form of arrays dimensioned by regions, years, ages, sexes and fleets, with internal validation to ensure compatibility with model dimensions. Optional data usage flags are also provided to allow users control over datasets (and years) that contribute to the likelihood (e.g., to accommodate not fitting years with missing data or non-representative sampling). Implementation of these helper functions returns a list object, which can then be used directly within RTMB to fit the specified model and perform standard non-linear function minimization. Once the model is optimized, SPoRC output helper functions then parse the RTMB objects and allow the user to complete the end-to-end assessment process (e.g., perform catch projections). For instance, extensive diagnostic, visualization and simulation functions can then be applied to an RTMB fitted model object to evaluate model performance and facilitate simulation-based model conditioning.

2.2.2 | Population Structure, Recruitment and Demographics

Primarily, SPoRC can model a single population (and single species) across an 'unlimited' number of regions, with constraints determined by the available data (e.g., spatial scale of reporting) and computational power. Population structure in SPoRC can be parameterized in several ways, depending on how spatial regions are coupled with recruitment assumptions (see Goethel et al. 2023 for additional descriptions of different population structures for assessment models). A panmictic population can be modelled with a single-region model using a stock-recruitment function that assumes global recruitment (density-dependent or mean) with annual deviations. A spatially heterogeneous population can also be modelled, in which a single population is distributed across multiple regions, potentially

TABLE 1 | Description of primary helper functions used to facilitate specification of a SPoRC model.

Function name	Model process	Description
Setup_Mod_Dim	Model dimensions	Initializes model dimensions, including regions, years, ages, lengths, sexes and fleets.
Setup_Mod_Rec	Recruitment	Determines recruitment processes and associated model options.
Setup_Mod_Biologicals	Biological	Specifies weight-at-age, maturity-at-age, ageing error, age-length transition matrices and natural mortality options.
Setup_Mod_Movement	Movement	Defines movement between regions and associated model options.
Setup_Mod_Tagging	Tagging	Configures tagging data usage, tag reporting rates and associated tagging submodel options.
Setup_Mod_Catch_and_F	Fishery removals and fishing mortality	Specifies observed catches, catch aggregation types, data usage flags, fishing mortality penalties and catch uncertainty.
Setup_Mod_FishIdx_and_Comps	Fishery indices and compositions	Configures fishery indices, age and length compositions, sample sizes, likelihoods, aggregation structures, and data usage flags.
Setup_Mod_SrvIdx_and_Comps	Survey indices and compositions	Configures survey indices, age and length compositions, sample sizes, likelihoods, aggregation structures and data usage flags.
Setup_Mod_Fishsel_and_Q	Fishery selectivity and catchability	Specifies fishery selectivity models, catchability estimation and priors.
Setup_Mod_Srvsel_and_Q	Survey selectivity and catchability	Specifies survey selectivity models, catchability estimation and priors.
Setup_Mod_Weighting	Data weighting	Determines likelihood weights for data sources and penalties to control their influence on likelihood contributions

with movement among them, and global recruitment assumed with apportionment estimated or specified to each region. Metapopulation structure can be represented by modelling multiple sub-populations, each with a unique localized density-dependent stock-recruitment function, with potential reproductive straying among sub-populations. Currently, SPoRC does not integrate functionality to represent natal homing dynamics and associated genetically-based demographic rates are not supported. Nonetheless, the generalized design allows a range of population and spatial structures to be readily explored within a single modelling framework.

Regardless of population structure assumption, demographic rates can be specified as either region-specific or region-invariant, allowing phenotypically-based differences to be modelled. Demographic parameters can be associated with a given region, such that a new demographic regime is adopted once an individual moves into a different region, similar to CASAL2, GADGET and VPA-2Box (Begley and Howell 2004; Doonan et al. 2016; Porch 2018). Both natural mortality and recruitment sex-ratios can be estimated and structured across relevant model partitions and parameterized as discrete blocks. While processes such as growth (i.e., weight-at-age and age-length transitions) and maturity-at-age cannot currently be estimated within SPoRC, externally derived values can be incorporated to describe population dynamics and observation processes (Table 2). This approach is common practice in many stock assessments, including Alaska sablefish for which SPoRC was initially designed for. However, we acknowledge that relying on external growth parameters may inadequately account for length-stratified sampling designs, size-selective mortality, and propagation of assessment uncertainty (Lee et al. 2024). Nonetheless, we anticipate that the modular design of SPoRC, coupled with RTMB's flexibility, can readily facilitate the future implementation of internal growth estimation or other extensions.

2.2.3 | Connectivity

When multiple regions are included, connectivity among regions are modelled as Markovian box-transfer dynamics. Several options for modelling connectivity are provided in SPoRC. In the simplest case, analysts can assume that no post-settlement movement occurs (i.e., an identity matrix). Alternatively, analysts can also assume that movement dynamics are known and provide an externally derived movement matrix. Priors following a Dirichlet distribution can

TABLE 2 | Summary of data types that can be integrated in SPoRC. For further details on the mathematical formulations of statistical likelihoods, refer to Data S1.

Data type	Statistical likelihood	Model treatment	Description
Fishery removals	Lognormal	Fit with statistical likelihoods	Time series of removals (weight or numbers) by a fishery fleet
Indices of abundance (or biomass; fishery and survey)	Lognormal	Fit with statistical likelihoods	Observed abundance or biomass by fishery or survey
Age and/or length compositions (Fishery and survey)	Multinomial (available with Francis re-weighting), Dirichlet-multinomial, and Logistic-normal	Fit with statistical likelihoods	Counts or proportions at age or length from fishery or survey
Tagging	Poisson, Negative Binomial, Multinomial (Release and recapture-conditioned), Dirichlet-Multinomial (Release and recapture-conditioned)	Fit with statistical likelihoods	Counts of individuals (region, age, and sex) recaptured by both fishery and survey fleets
Age-length transition matrix	—	Pre-specified input (assumed known)	Transition matrix to convert ages to lengths
Survey environmental covariates	—	Pre-specified input (assumed known)	Environmental covariates used to model temporal changes in survey catchability
Ageing Error	—	Pre-specified input (assumed known)	Ageing error used to fit to age-composition data
Weight-at-age	—	Pre-specified input (assumed known)	Weight-at-age during the spawning season, fishery operations, and/or survey operations
Maturity-at-age	—	Pre-specified input (assumed known)	Maturity-at-age to compute spawning stock biomass

also be provided to constrain movement estimation. In information-rich scenarios (i.e., integration of tagging and/or high-quality composition data across regions), movement dynamics could potentially be estimated across years, ages and sexes, with the simplest case assuming constant movement across these dimensions. Typically, informed parameter blocking would be used to balance parsimonious (and feasible) estimation against necessary complexity to adequately represent primary facets of connectivity that influence spatial population dynamics. Further flexibility is provided by allowing movement parameters to vary as independent deviations for each region across years, ages, and sexes, although computational challenges could potentially emerge (i.e., memory limitations) under this approach, especially as the number of model partitions increase. Alternative approaches to reduce the complexity of movement parameterization (e.g., functional forms and coarse-scale diffusion-taxis movement processes; Thorson et al. 2021) are planned for future development.

2.2.4 | Fishery and Survey Processes (Fleet Structure, Selectivity and Catchability)

SPoRC provides flexible specification of fishery and survey fleets, which is critical for representing removals and observations by age and across space. Users can define any number of fleets, constrained by data availability and the balance between model complexity and parsimony. Multiple fleets may operate within a region, and a single fleet can span all regions through shared parameters. Spatial processes can also be represented implicitly in single-region models specifying fleets-as-areas models (Berger et al. 2012). Selectivity can be modelled on either age or length (via an age–length transition matrix) using a range of functional forms. Selectivity may be time-invariant, vary as parametric blocks, or evolve through independent deviations or random walks. SPoRC also supports semi-parametric, time-varying selectivity by estimating correlations across years, ages, and/or cohorts using Gaussian Markov random fields (Cheng et al. 2023; Xu et al. 2019; Data S1). Catchability for fishery fleets can be configured to vary across space and time via discrete blocks, and survey catchability can be specified similarly, with the option to model continuous temporal variation as linear combinations of covariates.

2.2.5 | Data Integration and Observation Processes

The SPoRC platform allows statistical integration of multiple data sources (Table 2). Primary data sources for integrated age-structured assessment models like SPoRC include catch data, abundance indices, and composition data (age or length). Catch data (assumed to be lognormally distributed) inform population removals and scale, and, within SPoRC, can be incorporated as region- and/or fleet-specific catches. Abundance indices from fishery or survey fleets provide information on population scale and trends over time, and are fit assuming a lognormal likelihood. Removals-at-age, cohort strength, and population age structure are primarily informed by composition data. SPoRC allows fitting age and/or length compositions, but fitting length data requires the additional specification of an externally derived age-length transition matrix to convert predicted fishery or survey ages to lengths (Table 2). While SPoRC supports modelling region- and sex-specific partitions, composition data are sometimes observed on a coarser scale and functionality is provided to either fit aggregated (across both regions and sexes) or disaggregated compositions by partition and distinct time periods. Additionally, SPoRC includes composition error structures and data-weighting methods (Table 2) to ensure composition data are 'right-weighted' to balance their contribution relative to other data sources. For instance, Francis-reweighting can be applied using a helper function (e.g., when a multinomial likelihood is used) or self-weighting likelihoods can be utilized (Francis 2014). SPoRC can also incorporate covariates (as linear or non-linear effects) to estimate temporal changes in survey catchability. However, unlike other state-space assessment platforms, these covariates are assumed known, with no associated observation error (Stock and Miller 2021).

Tagging data can be integrated in SPoRC through a Brownie tag-attrition submodel that assumes dynamics similar to untagged population (Brownie and Robson 1983). Tagged cohorts are tracked by release region, age, and sex for each year at liberty. A primary assumption of the tagging sub-model is that the age of tagged fish is known (e.g., either derived from an age-length key or based on direct ageing). While assuming the age of tagged fish is known can be a strong assumption when ages are derived externally from the assessment model using an age-length key, the purpose of this assumption is to approximate mortality-at-age dynamics of the tagged population and to avoid the more restrictive alternative that all individuals experience the same mortality rate. Although some assessment models instead assume lengths are known and internally convert lengths to age-specific dynamics (either using an input age-length key within the model or estimating growth internally), we did not adopt this approach because repeatedly converting between length and age within the model can lead to 'smearing' of length frequencies, given that growth is modelled as length conditional on age. Nevertheless, the sensitivity of such an assumption can be evaluated by bootstrapping the age-length key, and improved methods to propagate age-length uncertainty within tagging models remains an active area of research. Tagged individuals are considered a separate model partition, but parameters are shared with the untagged population. Expected tag recaptures are derived using a modified version of Baranov's catch equation (Data S1). Modelling tag data requires the specification of several tag 'nuisance' processes to ensure that tagged individuals are representative of the untagged population. SPoRC allows users to specific a tag mixing period (Berger et al. 2024) so the remaining tagged population can reasonably represent the untagged portion of the stock. Options are also provided to fix or estimate tag induced mortality and tag shedding. Tag reporting rates can be specified (or estimated) and can be time-varying (with time blocks) and/or regionally varying. For fitting tagging data, several error structure options are provided (Table 2), with some structures able to be formulated as either release- or recapture-conditioned (McGarvey and Feenstra 2002).

2.3 | Parameter Estimation and Random Effects

The implementation of SPoRC in RTMB allows parameter estimation to occur using both frequentist and Bayesian paradigms. Fundamentally, SPoRC leverages Automatic Differentiation capabilities from RTMB to compute derivatives (first, second, and third) of a specified objective function (i.e., sum of likelihood contributions), which are then used in a gradient-based quasi-Newton optimizer to perform minimization via maximum likelihood estimation (Kristensen et al. 2016). Both penalized maximum likelihood and Laplace Approximation can be utilized to estimate parameter deviations (e.g., recruitment, selectivity). Penalized maximum likelihood is implemented through standard maximum likelihood operations, while Laplace Approximation implements a Taylor series expansion around the conditional modes, producing a multivariate normal approximation to the multi-dimensional integrals over parameter deviations to compute the marginal likelihood. Considering the complexity of age-structured stock assessment models, penalties (or priors in a Bayesian paradigm) can be specified on various parameters (e.g., natural mortality, catchability, movement) to constrain the estimation process. Uncertainty of parameter estimates can be approximated with the generalized delta method (Kristensen et al. 2016).

In some cases, both frequentist and Bayesian paradigms are jointly implemented in stock assessment models. Bayesian methods can provide a more complete characterization of uncertainty by fully exploring posterior distributions, which makes them valuable for management applications, as well as diagnostic checks for models estimated using maximum likelihood. Thus, SPoRC can also be implemented in a Bayesian framework, wherein both fixed and random effects are estimated using Markov Chain Monte Carlo (MCMC) via the No-U-Turn-Sampler (NUTS) algorithm. Support of Bayesian estimation is provided through R packages that interface RTMB and TMB model objects with the Stan software (Carpenter et al. 2017; Monnahan and Kristensen 2018).

2.3.1 | Reference Points and Projections

Short-term population projections are the final step in the end-to-end assessment process by providing the direct linkage between the assessment estimate of stock status and the information required for managers to provide catch advice. Thus, SPoRC integrates a module for estimating management reference points and conducting catch projections. For single-region applications, commonly utilized equilibrium biological reference points (BRPs) used for management, including F_{MSY} (fishing mortality associated with Maximum Sustainable Yield) and $F_{x\%}$ (fishing mortality associated with reducing spawning stock biomass per-recruit (*SSBPR*) to $x\%$ based on the spawning potential ratio, *SPR*) and its biological analogs can be readily computed (e.g., $B_{40\%}$ is the spawning stock biomass resulting from fishing at $F_{40\%}$, which is calculated as *SSBPR* multiplied by mean recruitment). Calculations of dynamic B_0 (unfished biomass under prevailing conditions) are also included. For spatially-explicit models, SPoRC can calculate both and values independently for each region (assuming no movement) or estimate global (population-wide) values, which account for movement, the distribution of recruitment, and regional fishing mortality-at-age (i.e., global reference points are essentially a weighted average of regional values, similar to the approach used in WHAM; Miller et al. 2025). Regional fishing mortality reference points that fully account for spatial dynamics (e.g., movement) can also be estimated when assuming region-specific density-dependent Beverton-Holt recruitment (i.e., metapopulation dynamics) following the methods of (Kapur et al. 2021). Additional details on calculations of single-region and spatial reference points can be found in Data S1.

Population projections can then be conducted under varying management scenarios, which generally follow the same population dynamics and model partitions as specified in a given assessment model. The projections can be deterministic or stochastic depending on how recruitment is simulated. In particular, recruitment processes can be specified as the mean of a time-series (e.g., full length or a portion of the time-series), to follow a deterministic Beverton-Holt relationship, be set to zero (i.e., no recruitment in the projection period), or be simulated using an inverse Gaussian distribution parameterized with historical recruitment estimates. Recruitment drawn from an inverse Gaussian is parameterized using historical recruitment estimates, where the arithmetic and harmonic means of the selected recruitment period are used to derive the parameters of the distribution. Because the distribution parameters are explicitly derived from recruitment estimates, stochastic projections are conditioned on the estimated mean and variability, rather than imposed through a naive log-scale assumption, as would be done in a lognormal formulation. Additionally, when the estimation model includes multiple fleets, fleet-specific projection scenarios can be specified. For instance, the allocation of fishing mortality to a given fleet can be adjusted to reflect anticipated changes in fishery fleet structure, or to account for scenarios in which catches from a given fleet are not fully utilized (e.g., Nielsen et al. 2021). Moreover, a variety of options are allowed for specifying how time-dependent processes vary in projections, where analysts can define their own assumptions regarding future variation in demographic parameters (e.g., natural mortality, selectivity, weight-at-age). Available options include projecting time-dependent processes forward with a specified correlation structure (e.g., autoregressive processes), fixing parameters at terminal year estimates, or applying multi-year averages. The projection module also allows flexible definition of harvest control rules, and thus can be readily modified to facilitate compliance with a variety of national mandates for providing harvest advice.

2.3.2 | Model Diagnostics

Model diagnostics and output graphics are not commonly built directly into assessment platforms, often requiring the user to export model outputs to a separate programming language and/or package (Taylor et al. 2021). SPoRC provides a suite of tools to facilitate efficient model evaluation. Standard convergence diagnostics, including maximum absolute gradients, large parameter standard errors or correlations among parameters, and verification of a positive definite Hessian matrix are readily provided to analysts. Through various helper functions (Table in Data S2), model estimates can be visualized, with the ability for post hoc adjustments to default graphics via integration with ggplot2 (Wickham 2016). Additional model diagnostic features include retrospective analyses, runs tests on Pearson residuals (Carvalho et al. 2021; Wald and Wolfowitz 1940), likelihood profiles, jitter analysis, one-step ahead residuals for composition data (R package compResidual; Trijoulet et al. 2023) and Bayesian MCMC diagnostics (R package adnuts and tmbstan; Monnahan and Kristensen 2018). Further demonstration of using these diagnostic tools within the SPoRC platform can be found at https://chengmatt.github.io/SPoRC/articles/o_get_started.html.

2.3.3 | Simulation Features

Simulation-based analyses are critical for evaluating the robustness of assessment and management processes (Punt et al. 2016). SPoRC supports three complementary simulation approaches: self-testing, cross-testing and closed-loop simulations for MSE. These simulation features allow analysts to define operating models to reflect user-specified population dynamics, generate pseudo datasets with varying levels of observation and process error, and evaluate the performance of estimation models under known simulation conditions, both when model assumptions are correct (self-test) and when they differ from the simulated truth (cross-test). Self-tests are useful for assessing potential issues pertaining to model identifiability given the available data, while cross-tests can illustrate the implications of model misspecification (e.g., incorrectly assuming constant selectivity when it varies over time). SPoRC also supports closed-loop simulations (i.e., MSE), which additionally includes feedback between the management procedure (MP; the combination of data observations, assessment, and harvest control rule) and the population dynamics through forward simulation of specified catch from the system based on the MP. MSE allows testing the robustness of the entire assessment-management process, while comparing alternative MPs through evaluation of trade-offs in specified performance metrics (e.g., fishery and conservation-based outputs). In general, analysts have considerable flexibility to specify how the realized catches relate to stock status and socio-economic considerations. For instance, catch may not be assumed to be fully utilized in each evaluation period of the MSE, allowing for partial harvest or additional catch variability due to compliance, market forces, or other factors. Moreover, a generalized number of fishery fleets can be incorporated into SPoRC, allowing for simulation testing of MPs under different fleet-specific behaviours and quota allocations. Development of simulations is facilitated through helper functions (Table in Data S2), where the user provides specific inputs for the operating model (e.g., biology or fishery dynamics), management options (i.e., for MSE) or an RTMB object to condition simulations based on an assessment model.

2.4 | Case-Study Demonstrations

In the following, we present three case-studies encompassing a range of life-history types and data availability to demonstrate SPoRC's flexibility in representing spatial structure, ability to

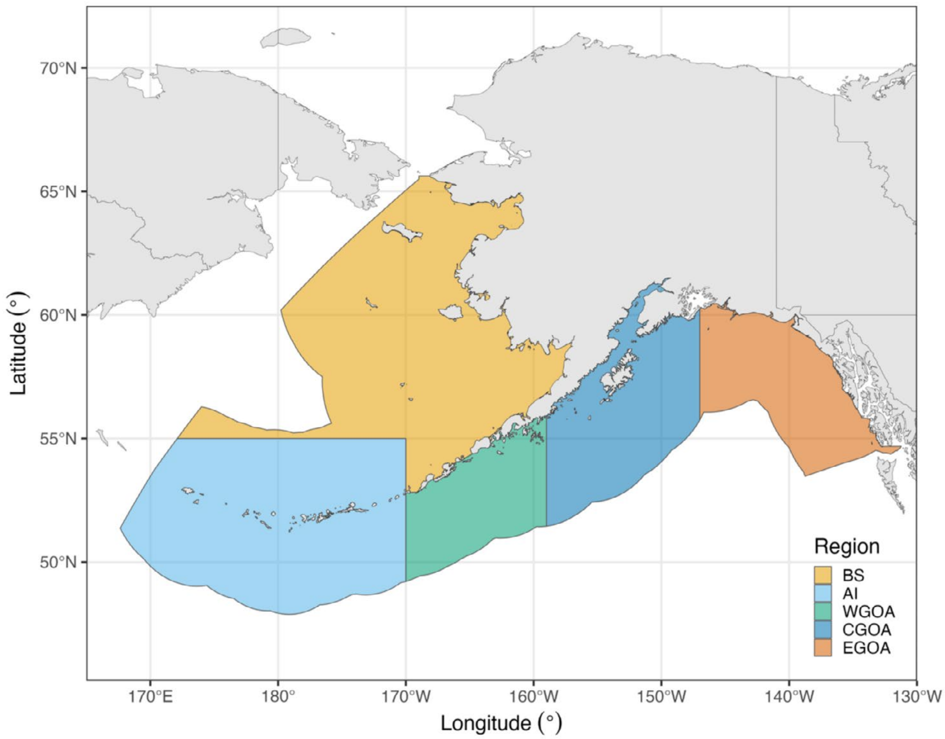

FIGURE 1 | Map of Alaska depicting the regions (coloured polygons) included in the three case-studies presented. Regions illustrated for the spatial models include the: Bering Sea (BS), Aleutian Islands (AI), Western Gulf of Alaska (WGOA), Central Gulf of Alaska (CGOA) and Eastern Gulf of Alaska (EGOA).

specify alternative random effect structures, and its utility in conducting simulation-based model evaluations. Approximate model run-times for each case-study are reported to provide readers with a sense of computational requirements and expectations when applying SPoRC to their own data.

2.4.1 | *Regions* Case-Study

In the first case-study (*Regions*), we apply single-, three- and five-region models for Alaska sablefish. Spawning stock biomass (*SSB*), recruitment estimates, and population projections are compared across models to illustrate inference that can be gained through simultaneous application of models assuming different spatial structures. The single-region model pooled data across the Bering Sea (BS), Aleutian Islands (AI), Western Gulf of Alaska (WGOA), Central Gulf of Alaska (CGOA) and the Eastern Gulf of Alaska (EGOA) (Figure 1). The three-region model considered the BS, AI and WGOA as one region, while treating the CGOA and EGOA separately. The five-region model treated each region distinctly. All models closely followed the operational stock assessment model, with two sexes, 30 ages and two fishery fleets (Goethel and Cheng 2024), while fitting to catch, age and length compositions, and abundance index data. The three-and five-region models additionally used tagging data and included three movement age blocks (i.e., ages 2–7, 8–15 and 16–31) to reflect ontogenetic patterns (Cheng, Goethel, et al. 2025; Cheng, Marsh, et al. 2025). All models used in the *Regions* case-study were fit to real world data.

Fishing mortality reference points for population projections were all calculated at the population (global) scale and were derived based on 40% of the unfished aggregate *SSBPR*, providing estimates of $F_{40\%}$ and associated $B_{40\%}$. For the spatial models, *SPR* reference points explicitly accounted for movement, recruitment apportionment, and regional differences in fishing mortality-at-age (Data S1). Thus, $F_{40\%}$ in the spatial models were estimated and applied at the region-wide scale. However, because mean recruitment can be defined spatially, region-specific values of $B_{40\%}$ were derived (sum of region-specific mean recruitment multiplied by regional *SSBPR*). Thirty-year deterministic projections (assuming region-specific mean recruitment) were conducted by applying a ramped threshold control rule, with global $F_{40\%}$ as the maximum fishing mortality rate and region-specific $B_{40\%}$ as the target biomass (Figure 2). Under the rule, the maximum fishing mortality rate applied to a region declines linearly as region-specific *SSB* falls below its associated $B_{40\%}$target. In the spatial context, applying a global $F_{40\%}$ achieves the global $B_{40\%}$ (sum of region-specific $B_{40\%}$), but has the potential to not necessarily attain the region-specific targets due to movement among regions and corresponding differences in target mortality (i.e., due to region-specific control rules, despite the same overarching global $F_{40\%}$ target value).

2.4.2 | Random Effects Case-Study

In the second case-study (*Random Effects*), we demonstrate how continuous time-varying fishery selectivity can be specified as random effects using Eastern Bering Sea walleye pollock. The model configuration closely follows the operational assessment, employing a single-region, single-sex structure, with 15 modelled ages (Ianelli et al. 2024). One fishery fleet and

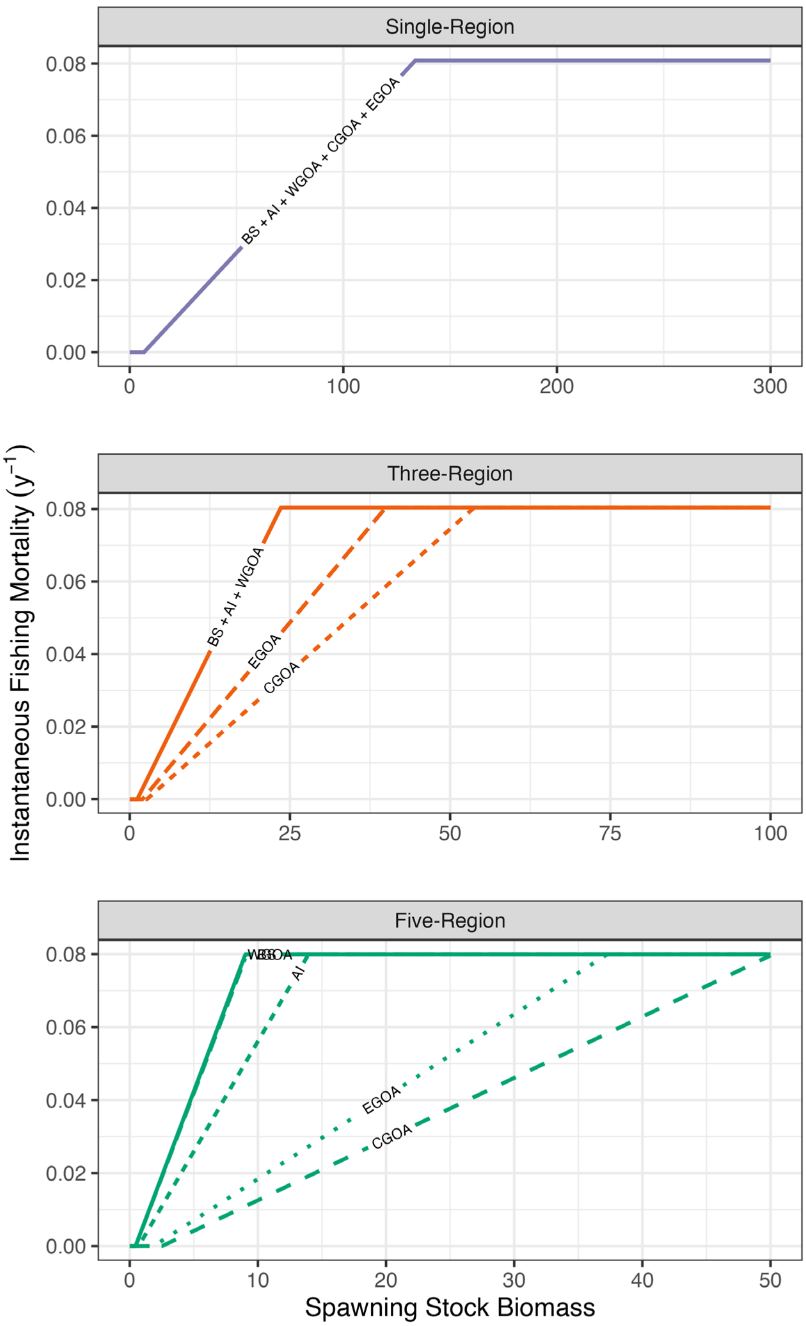

FIGURE 2 | Illustration of how fishing mortality rates change with region-specific spawning stock biomass (*SSB*) under the ramped threshold control rule for the *Regions* case-study. Line types and associated letters (Bering Sea (BS), Aleutian Islands (AI), Western Gulf of Alaska (WGOA), Central Gulf of Alaska (CGOA), Eastern Gulf of Alaska (EGOA)) indicate control rules applied for a given region. For spatial models (Three-Region and Five-Region), the biomass level at which the ramp begins differs across regions due to the ability to define region-specific quantities. Because a global *SPR* reference point is used, the estimated fishing mortality reference point is identical across regions.

three survey fleets are included, with real world fishery catch and age compositions, along with survey indices and age compositions jointly fitted. In the operational assessment, time-varying fishery selectivity is modelled non-parametrically with random-walk deviations. Bottom trawl survey selectivity is modelled as a logistic function with random-walk deviations and an additional age-one deviation. For both acoustic survey fleets, selectivity is modelled non-parametrically with random-walk deviations. Given the aforementioned complexity, and to enable estimation of fishery selectivity as random effects, selectivity for all surveys was held constant (simplification did not appear to substantially impact model interpretation). Thus, for the *Random Effects* case-study, fishery selectivity was modelled as a logistic function, and a total of 6 alternative model structures were evaluated. The models evaluated included: (1) constant selectivity, which serves as a baseline for comparison (*constant*), (2) independent and identically distributed (iid) parametric deviations (*iid_p*; parameters describing the age-at-50% selection and slope each vary independently), (3) random-walk parametric deviations (*rw_p*;

parameters describing the age-at-50% selection and slope each vary as a random walk), (4) iid semi-parametric deviations (*iid_sp*), (5) two-dimensional autoregressive semi-parametric deviations with correlations across ages and years (i.e., *2dar1_sp*) and (6) three-dimensional autoregressive semi-parametric deviations with correlations across ages, years and cohorts (*3dgmrf_sp*) (Table 3). For parametric deviations, deviations were applied to the logistic selectivity parameters, whereas, for semi-parametric derivations, variations were applied directly to the functional form allowing for departures from the logistic form.

2.4.3 | *Desk MSE* Case-Study

The third case-study ('*Desk MSE*') implemented a desktop MSE for Gulf of Alaska Dusky rockfish to examine trade-offs among four alternative management procedures (MPs) under two recruitment scenarios (Table 3). Briefly, the MSE consisted of operating models conditioned on stock assessment model estimates from 1977 to 2024 (Omori et al. 2024), followed by a 50-year feedback simulation period. Operating models generated pseudo-datasets every 2 years to represent data collection processes, and estimation models based on a SPoRC implementation of the operational stock assessment were fit to simulated datasets at the same interval. Estimates from the estimation model were used to project prescribed catches 2 years ahead, which were assumed to be fully utilized by the fishery. This process was then repeated across 200 simulations. Typically, a full MSE would include engagement with stakeholders to develop MPs and associated performance metrics. Because our study does not incorporate stakeholder input, we include the term 'desktop' to emphasize the exploratory nature of the exercise (Walter III et al. 2023).

The MPs evaluated include (following similar methods from Zahner et al. 2025):

1. *f0*: No fishing, serving as a baseline for comparison,
2. *f40_thresh*: A threshold control rule using $F_{40\%}$ and $B_{40\%}$ as reference points. Fishing mortality is set at $F_{40\%}$ when $SSB \geq B_{40\%}$ and declines linearly as SSB falls below $B_{40\%}$. No fishing is allowed if $SSB \leq B_{5\%}$. This rule represents the default strategy used for Gulf of Alaska Dusky rockfish management (Omori et al. 2024).
3. *f40_const*: A constant fishing mortality rate set at $F_{40\%}$ regardless of *SSB*.
4. *f40_hybrid*: A hybrid rule combining *f40_thresh*, a higher biological reference point ($B_{60\%}$) in place of $B_{40\%}$, and a harvest cap (3000 tons, defined based on average historical catches).

To assess trade-offs among MPs, two recruitment scenarios were simulated. In the *rand* scenario, recruitment was randomly resampled from the historical time-series to mimic past recruitment dynamics. In the *crash* scenario, recruitment followed Beverton-Holt dynamics and was assumed to decline sharply to mimic environmentally induced recruitment collapses. During the first 23 years of the simulation period (2025–2046), recruitment was simulated under low productivity conditions (virgin recruitment, $R_0 = 2.5$; steepness, $h = 0.5$), followed by a recovery phase (2047–2074) with higher productivity ($R_0 = 12$; $h = 0.75$). Recruitment parameter values were selected to broadly reflect realistic bounds, with the higher productivity phase generally approximating the mean of the historical recruitment time series. Metrics to assess MP performance included *SSB*, catch, and catch average annual variation (AAV), summarized across simulation replicates as the median with 95% simulation intervals.

3 | Results

The suite of models evaluated for the *Regions* case-study successfully converged, producing positive definite Hessian matrices and small maximum absolute parameter gradients. Model run-times varied, with the single-region model completing in seconds, the three-region model in approximately 30 min, and the five-region model in roughly an hour. In general, these models demonstrated similar global *SSB* scale and trend estimates (Figure 3A), though a slightly higher scale was estimated for the single-region model. Estimates of $F_{40\%}$ were nearly identical across models (ranging from 0.079 to 0.080), while global $B_{40\%}$ estimates were slightly higher for the single-region model. Projections of global *SSB* equilibrated at global $B_{40\%}$ for the single-region model, and at both the global and regional biomass reference point for the spatial models. While estimates of global scale and trend were broadly consistent between models, spatial models revealed region-specific differences in biomass and recruitment, with the GOA demonstrating the highest spawning stock biomass levels and western regions exhibiting strong age-2 recruitment (Figure 3B and 3C, Figure in Data S2).

Across the fishery selectivity model formulations evaluated in the *Random Effects* case-study, model run-times were fastest for *constant* (seconds), followed by models assuming parametric deviations (*iid_p* and *rw_p*; ~15 min). Models assuming semi-parametric deviations were often the slowest (*iid_sp*, *2dar_sp*, *3dgmrf_sp*; ~30 min). In general, estimates of recruitment and *SSB* were consistent across models, with differences largely emerging during the early period and towards the end of the time-series (Figure 4A). These differences likely stem from the hierarchical structures specified for selectivity random effects, which define how information is shared across adjacent years or age groups when data are sparse (i.e., as is the case early and late in the time series when limited composition data are available for a given cohort), and whether the configuration was specified as parametric (*constant*, *iid_p*, and *rw_p*) or semi-parametric (*iid_sp*, *2dar1_sp*, *3dgmrf_sp*). For example, the *2dar1_sp* and *3dgmrf_sp* models incorporated correlations in the random effects structure, likely allowing information from neighbouring groups to propagate into data-sparse groups (i.e., partial pooling or 'shrinkage'), supporting stronger deviations away from a zero mean (Figure 4B). By contrast, models without correlated random effects (i.e., *iid_sp*) tended to revert to a zero mean in data-sparse situations. Furthermore, parametric selectivity enforced a strict assumption that selectivity remained asymptotic across the entire time series, whereas semi-parametric models allowed selectivity to flexibly shift between asymptotic or dome-shaped patterns (Figure 4B).

TABLE 3 | Model descriptions for each application in each of the three case-studies.

Case study	Application species and region	Models/ Management procedures	Purpose	Related code implementations
Regions	Alaska sablefish	*Single-Region*, *Three-Region*, *Five-Region*	Demonstrates flexibility in transitioning between single-, three-, and five-region models, and highlights spatial structure effects on population dynamics.	https://chengmatt.github.io/SPoRC/articles/i_reference_points.html#reference-points https://chengmatt.github.io/SPoRC/articles/g_spatial_sablefish_case_study.html
Random Effects	Eastern Bering Sea walleye pollock	*constant*, *iid_p*, *rw_p*, *iid_sp*, *2dar1_sp*, *3dgmrf_sp*	Illustrates the implementation of annually-varying fishery selectivity as random effects.	https://chengmatt.github.io/SPoRC/articles/f_single_region_ebs_pollock_case_study.html https://chengmatt.github.io/SPoRC/articles/n_single_region_ebs_pollock_randomeff_case_study.html
Desk MSE	Gulf of Alaska Dusky rockfish	*f0*, *f40_thresh*, *f40_const*, *f40_thresh_cap*	Implements a desktop MSE to examine trade-offs among management procedures across recruitment scenarios (*rand* and *crash*).	https://chengmatt.github.io/SPoRC/articles/h_closed_loop_simulations.html https://chengmatt.github.io/SPoRC/articles/p_single_region_dusky_alt_mp_testing.html

For the Gulf of Alaska Dusky rockfish *Desk MSE* case-study, model run-times within a given assessment period were relatively fast, completing on the order of seconds. The case-study revealed several trade-offs in performance metrics among the evaluated MPs. Catch trajectories for the *rand* recruitment scenario remained stable across MPs, with lower harvest levels for MPs imposing harvest caps (*f40_hybrid*; Figure 5A). As expected, *SSB* trajectories were elevated for MPs that exerted lower harvest. Differences in catch AAV were most pronounced, with MPs implementing harvest caps (*f40_hybrid*) showing the greatest stability and *f40_thresh* showing the least. Under the *crash* recruitment scenario, larger differences in catch and *SSB* emerged (Figure 5A). MPs that utilized biomass reference points quickly reduced catch levels (most notably *f40_hybrid*) and maintained higher *SSB* throughout the time series. In contrast, the *f40_const* MP permitted higher catches during periods of reduced recruitment, but produced lower catch levels than other MPs following recruitment increases (Figure 5A). This pattern likely arose because elevated harvest during low-recruitment period depressed *SSB* levels, foregoing potential catch in subsequent years. Moreover, it was notable that the *f40_hybrid* MP provided more balanced trade-offs across both recruitment scenarios, maintaining improved *SSB* and intermediate catch levels, while achieving substantial improvements in catch AAV (Figure 5B).

4 | Discussion

Over the past several decades, the field of fisheries science has seen substantial quantitative advancements, with a 'Golden Age' between the 1980s and the 2000s marked by the development of statistical catch-at-age models (Quinn 2003). Since then, advances in computational methods, statistical approaches, novel data sources, and understanding of non-stationary dynamics have further improved stock assessment methodology. Keeping pace with these rapid developments remains challenging, highlighting the need for modular next-generation stock assessment platforms that can readily integrate new advances, while maintaining the essential features of existing tools.

4.1 | Addressing Next Generation Stock Assessment Priorities

SPoRC exemplifies a next-generation stock assessment platform designed to meet evolving management and research needs (Punt et al. 2020). It provides a modular, transparent, open-source framework implemented in R with C++ templates through RTMB, thereby, supporting open science practices through accessible code sharing and development. Moreover, SPoRC integrates extensive flexibility to address emerging complexities (e.g., spatial and sex-structured dynamics) to readily meet the needs of operational stock assessment models (e.g., Alaska sablefish). In particular, spatial model design requirements are met through the ability of users to choose among a range of spatial configurations, recruitment assumptions, fishery fleet structures, and connectivity parameterizations (Berger et al. 2024). Similarly, the modular design enables efficient scaling between single-region, fleets-as-areas and multi-region models, provided that data workflows are similarly adaptable (e.g., aggregation of multi-region inputs for single-region applications). SPoRC also enables implementation of an end-to-end assessment process within a unified platform by including diagnostic tools to evaluate model performance and robustness, rapid visualization of model outputs, estimation of reference points and projections to inform management advice, and simulation features for testing assessment-management frameworks. Collectively, these features position SPoRC as a flexible and accessible platform supporting the development of robust science-based fisheries management advice in increasingly dynamic ecosystems (Punt et al. 2020; Dichmont et al. 2025).

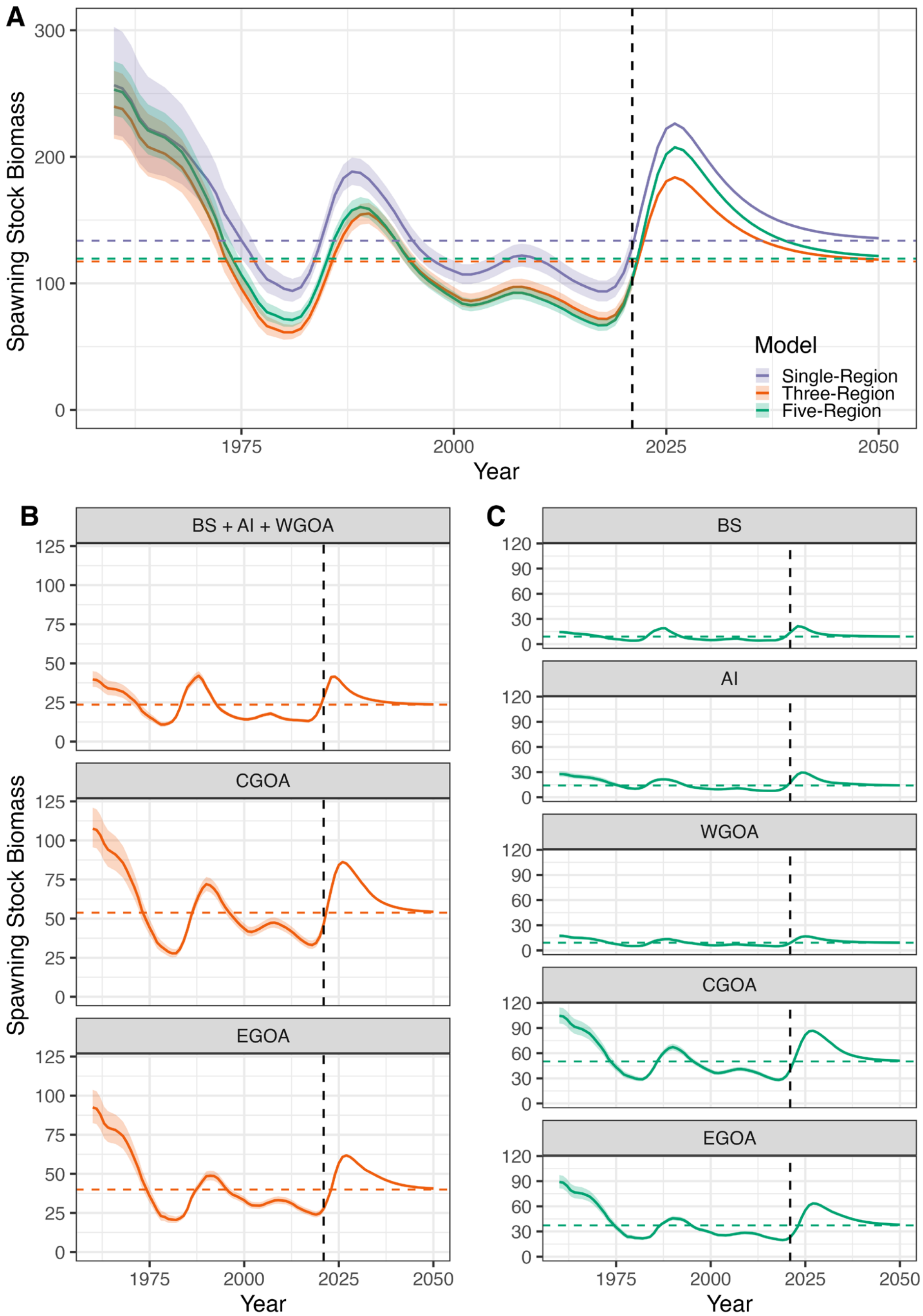

FIGURE 3 | Results of the application of single-region, three-region, and five-region model configurations for Alaska sablefish in the *Regions* case-study. In all panels, solid lines show spawning stock biomass (*SSB*) estimates and shaded regions indicate 95% confidence intervals. Dotted vertical lines represent the projection period and dashed horizontal lines represent either global (panel A) or local (panels B and C) $B_{40\%}$. Panel A depicts region-wide estimates, while panels B and C show region-specific estimates from the three-and five-region model, respectively. Regions depicted for the spatial models include the: Bering Sea (BS), Aleutian Islands (AI), Western Gulf of Alaska (WGOA), Central Gulf of Alaska (CGOA) and Eastern Gulf of Alaska (EGOA).

4.2 | Comparison of SPoRC to Existing Assessment Platforms

Although many of the existing primary assessment platforms can integrate some form of spatial structure, next generation successors are in development due to the challenges of existing platforms to adequately adapt rapidly evolving assessment good practices. For instance, CASAL2 and GADGET remain among the most flexible assessment platforms in terms of spatial capabilities (e.g., supporting nearly all common population structures). However, implementing random effects is challenging, options for composition and tagging likelihoods are relatively limited (i.e., limited to Multinomial or Dirichlet-multinomial likelihoods for compositions, lack of recaptured conditioned approaches; Berger et al. 2024), and simulation implementation is less straightforward. Likewise, while SS3, MULTIFAN-CL and VPA-2Box are among some of the more flexible assessment platforms, and are able to accommodate diverse population

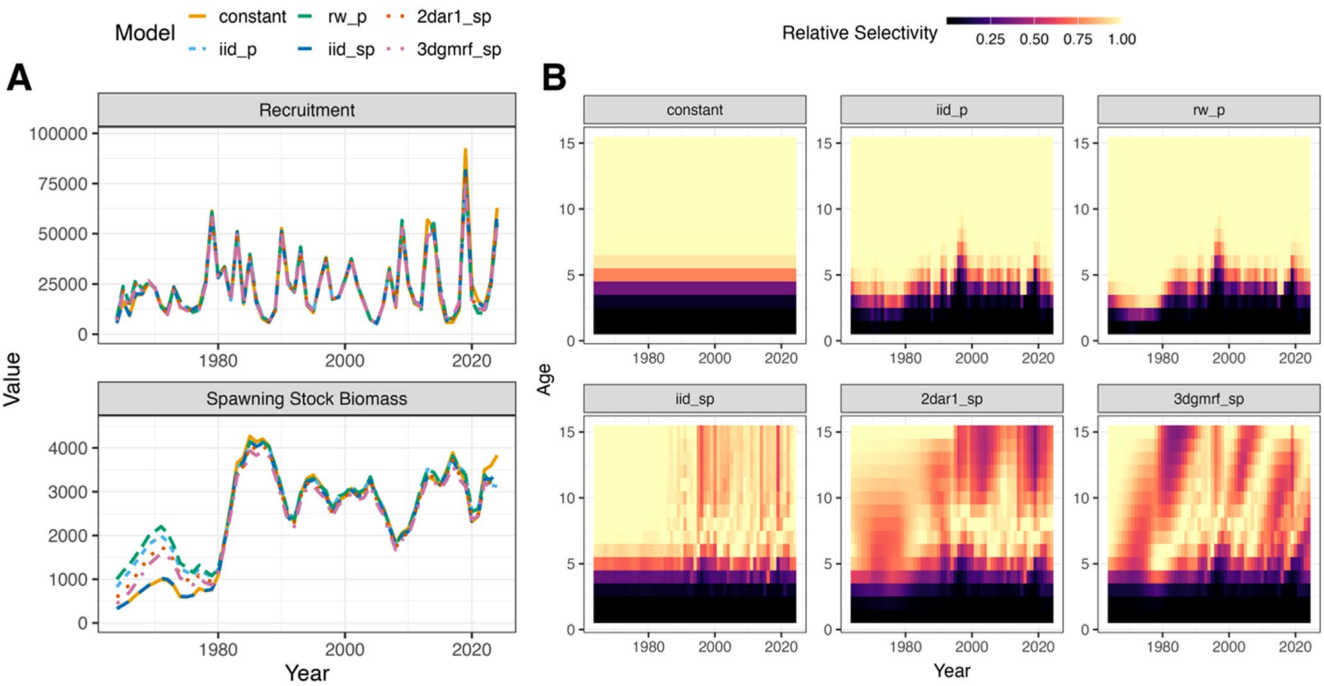

FIGURE 4 | Model results from the *Random Effects* case-study applied to Eastern Bering Sea walleye pollock. Panel A depicts recruitment and spawning stock biomass (*SSB*) trends estimated across the six fishery selectivity model formulations (colours and line type) evaluated. Panel B shows the estimated fishery selectivity for each model.

structures as well as many general assessment modelling features, they lack integration of random effects. In contrast, SAM provides a robust framework for implementing random effects, though it does not currently incorporate spatial nor sex-specific population dynamics (Nielsen and Berg 2014). While FIMS also supports random effects, development is still ongoing, with continued progress expected. The WHAM platform allows efficient integration of random effects, incorporates simulation features (natively and via whamMSE; https://github.com/lichengxue/whamMSE), and was recently extended to incorporate spatial capabilities. However, it is limited to natal-homing dynamics, cannot integrate tagging data and is also unable to accommodate sex-specific dynamics (Miller et al. 2025; Stock and Miller 2021).

Thus, SPoRC is currently positioned as one of the more flexible platforms that integrates spatial and sex-specific dynamics and random effects, thereby, bridging the capabilities of existing platforms (Table 4). In addition to its flexibility, SPoRC generally achieves shorter run times compared with most bespoke ADMB models and other existing assessment platforms; for example, SPoRC ran approximately twice as fast as the bespoke single-region ADMB sablefish model. Moreover, it is a fully unified package allowing the user to perform the entire assessment process internally, while also enabling straightforward simulation and MSE testing. In the coming years ongoing development is expected to further enhance SPoRC capabilities, especially to address key decision points in spatially-explicit stock assessment models. In particular, priority developments are to incorporate natal homing (which can mimic the growth 'morph' concept in SS3; Taylor and Methot 2013), genetically-based demographic rates and seasonality (Miller et al. 2025; Porch 2018). Additional features will also be introduced as needed to support other stock assessments and research models (e.g., linking fine-scale spatio-temporal models with stratified models; Thorson et al. 2021).

To ensure the long-term functionality and responsiveness of SPoRC, maintenance and user support are provided through active use by assessment scientists at the Alaska Fisheries Science Center and in collaboration with researchers at the University of Alaska Fairbanks (the authors of this study). Importantly, partial funding mechanisms are in place to support continued development. We recognize, however, that SPoRC is not a universal solution for all stock assessment model applications. Multi-species functionality is not a priority (Adams et al. 2025), strictly length-based models cannot be implemented, and age-length model dynamics are not currently implemented (as well as conditional-age-at-length data), which can be useful for accounting for size-selective mortality processes (i.e., the 'Rosa Lee phenomenon'; Punt et al. 2017; Taylor and Methot 2013). In a similar vein, growth-related parameters must currently be derived external to the assessment model, which may fail to adequately account for size-specific processes (Lee et al. 2024). Moreover, compared with alternative platforms such as WHAM, SPoRC's capacity to incorporate environmental covariates is less developed. Nonetheless, implementation through RTMB simplifies integration of new features from a broad potential user base, supports more efficient model testing and ensures that the platform aligns with advances in next-generation assessment models.

4.3 | Potential Advancements in Operational Assessments Using SPoRC

As with any assessment platform, SPoRC supports exploration of alternative model structures, while enhancing reproducible and team-oriented workflows (e.g., as outlined in the ICES Transparent Assessment Framework, TAF; https://github.com/ices-taf). Its capacity for simulations and ease of implementing models of varying

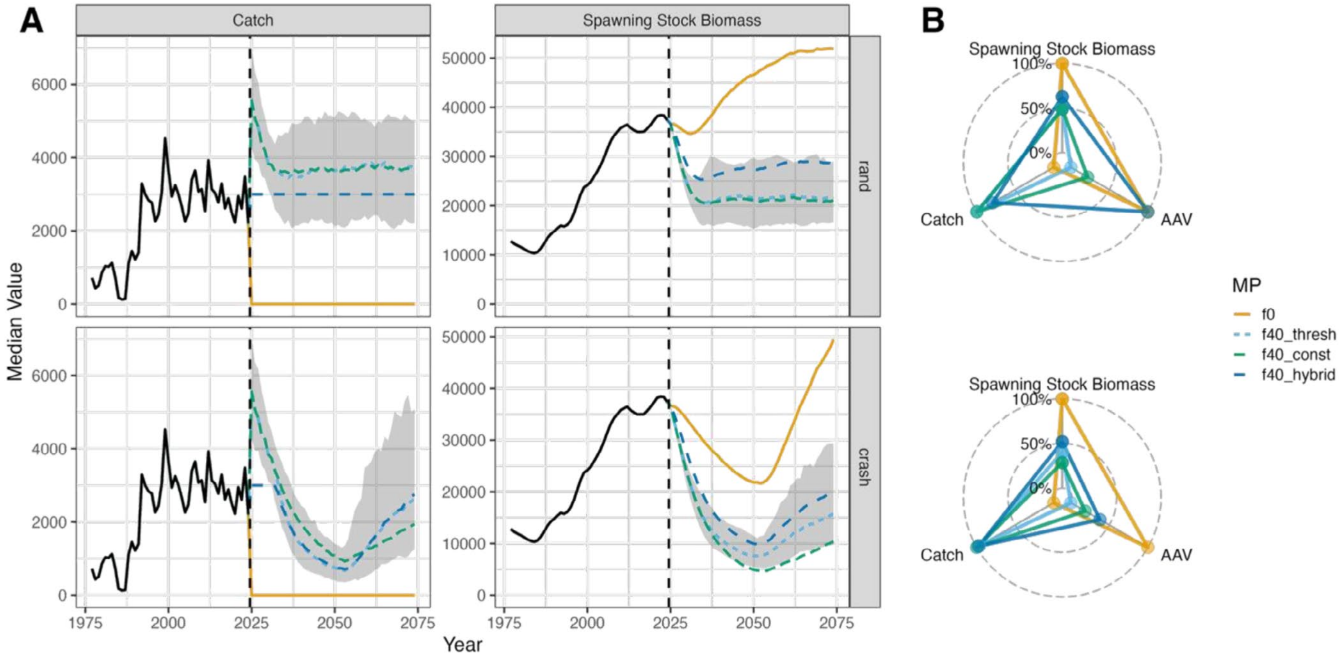

FIGURE 5 | Performance metrics from the *Desk MSE* case-study for Gulf of Alaska Dusky rockfish across the evaluated management procedures (MPs). Panel A shows median trajectories of catch and spawning stock biomass (*SSB*) across alternative MPs (colour and line type). Black solid lines represent the historical simulation-conditioned period, while coloured lines are simulated realizations from applying a specified MP. Note that catches increase rapidly in the early simulation period due to historical underutilization of the resource. The black vertical line marks the start of the simulation period. Grey shading shows simulation intervals for the *f40_thresh* rule and is provided for reference. Radar plots (Panel B) show trade-offs among MPs across key performance metrics. Values are scaled relative to the maximum median observed across MPs for each metric to facilitate comparison. For catch AAV, values are inverted such that higher values indicate better performance (i.e., lower variability). Panels in the first row depicts performance under the *rand* scenario, while the second row depicts performance under the *crash* recruitment scenario.

TABLE 4 | Primary features of SPoRC in comparison to capabilities of existing assessment platforms.

Process	Feature	Platforms with similar capabilities
Population structure	Panmictic, Spatially Heterogenous, and Metapopulation Structure	CASAL2, GADGET
Spatial structure	Any number of regions	CASAL2, GADGET, MULTIFAN-CL, SS3, WHAM
Stock-recruitment	Global and Local Density Dependence	CASAL2, GADGET
Fishery dynamics	Multiple fleets across regions and parameter sharing	CASAL2, GADGET, MULTIFAN-CL, SS3, WHAM, VPA-2Box
Demographics	Area-invariant or phenotypically-based	CASAL2, GADGET, SS3, WHAM, VPA-2Box
Data integration	Mark-recapture data	CASAL2, GADGET, MULTIFAN-CL, SS3, VPA-2Box
Random effects	Integration of random effects via Laplace Approximation	WHAM, SAM, FIMS
Simulation	Cohesive simulation testing (i.e., not an external package to the platform) via self-and cross-tests, and closed-loop functionality	WHAM, SAM, FIMS

complexity, particularly for spatial structures, helps identify appropriate model complexity and strengthens management inference by conveying multiple modelling perspectives.

Notably, the *Regions* and *Random Effects* case-studies illustrate SPoRC's ability to advance operational stock assessments within a unified framework. The *Regions* case-study demonstrated that SPoRC enables simultaneous implementation and updates to both single-region and multi-region models within an annual assessment cycle, with multi-region models providing insights not evident from single-region approaches. In particular, multi-region models suggested that Alaska sablefish likely undertake ontogenetic movement, with recruitment generally concentrated in western regions and individuals migrating eastward as they mature (Figure 3, Figure B1; Cheng, Marsh, et al. 2025), offering new understanding of localized population dynamics. Meanwhile, the *Random Effects* case-study underscored SPoRC's ability to efficiently explore a range of time-varying selectivity parameterizations in Eastern Bering Sea walleye pollock, highlighting that semi-parametric selectivity configurations potentially improve the characterization of removals-at-age by enabling departures from prespecified functional forms (Figure 4).

Building on these capabilities, SPoRC further advances operational assessment capacity by facilitating efficient exploration of management strategies, thus strengthening inference on the robustness of management frameworks. The *Desk MSE* case-study demonstrated SPoRC's ability to rapidly conduct closed-loop simulations for Gulf of Alaska Dusky rockfish (i.e., within a given assessment cycle), revealing key trade-offs in performance metrics among MPs. MPs without a biomass target reference point maintained consistent catch levels, but performed poorly during periods of low recruitment due to limited capacity to reduce harvest (Figure 5). In contrast, hybrid MPs that incorporated threshold rules with harvest caps and/or elevated biomass reference points better buffered against uncertainty in future recruitment dynamics, achieving more balanced outcomes across performance metrics. Thus, the capacity to conduct MSEs in a timely manner provides a valuable tool for evaluating the likely consequences of alternative management advice on population dynamics, supporting the provision of more effective fisheries management, particularly in situations where exceptional circumstances are declared (de Moor et al. 2022).

4.4 | Conclusions

With non-stationary changes in spatial and ecosystem dynamics, next-generation assessment platforms must be capable of addressing these considerations to provide effective management advice. Broader adoption of spatially-explicit assessment models therefore offer a crucial pathway for enhancing collective understanding of how spatial, temporal and ecosystem processes shape marine population dynamics. Accordingly, flexible, modular and cohesive assessment platforms are essential to overcoming institutional barriers in operationalizing spatial stock assessments, enabling knowledge sharing, model development and the expansion of and insular expert guild (Hilborn 2003; Goethel et al. 2024). Ultimately, platforms with relatively lower learning curves (i.e., through accessible interfaces, extensive package vignettes, and usability focused-design) can better engage scientists across an array of disciplines (e.g., ecosystem modellers, biologists, stock assessment scientists), supporting the development of tools and knowledge that cohesively link spatial and ecosystem processes and promote more sustainable, holistic fisheries management practices.

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Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

All analyses were conducted using the SPoRC package, release 1.0.0 (https://github.com/chengmatt/SPoRC). Additional vignettes associated with this manuscript can be found at https://chengmatt.github.io/SPoRC/articles/index.html.

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Supporting Information

Additional supporting information can be found online in the Supporting Information section. **Data S1:** faf70082-sup-0001-SupplementaryMaterialA.docx. **Data S2:** faf70082-sup-0002-SupplementaryMaterialB.docx.